TRƯỜNG ĐẠI HỌC QUỐC TẾ MIỀN ĐÔNG
EASTERN INTERNATIONAL UNIVERSITY

BECAMEX BUSINESS SCHOOL

Course: MIS 451 - Machine Learning for Business

GROUP PROJECT

PREDICTING ONLINE SHOPPERS' PURCHASING INTENTION

Lecturer: Ms. Huynh Gia Linh

Group: POLYGON

Student's Name	Student’s IRN
Nguyen Ngoc Minh Thu	2232300163
Nguyen Thi Hoai Thuong	2232300040
Tran Ngoc Trung Hieu	2132300543

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1. Business Context and Objectives

1.1 Background

In the e-commerce environment, understanding whether a website visitor is likely to complete a purchase is an important business problem. Online retailers collect a large amount of behavioral data during each browsing session, such as the number of pages visited, time spent on different page categories, bounce rate, exit rate, page value, traffic type, visitor type, and visit timing. These session-level variables can provide useful signals about a visitor's purchasing intention.

The Online Shoppers Purchasing Intention Dataset from the UCI Machine Learning Repository is suitable for this study because it contains real session-level data from an e-commerce website. The dataset includes 12,330 online shopping sessions and uses revenue as the target variable, where the objective is to predict whether a session ends with a purchase. Since only a minority of sessions result in revenue, the dataset reflects a common business challenge in online retail: most visitors browse without purchasing, while only a smaller group converts into paying customers (Sakar & Kastro, 2018).

From a business perspective, predicting purchasing intention can help online retailers identify potential buyers earlier and support more effective marketing decisions. For example, customers predicted to have high purchase intention can be targeted with personalized promotions, product recommendations, or customer support interventions. In contrast, customers with low purchase intention may require different engagement strategies to reduce website abandonment.

From an academic perspective, this problem is relevant because previous studies have shown that clickstream and session-level behavioral data can be used to predict online purchase behavior. Sakar et al. (2019) proposed a real-time online shopper behavior analysis system using pageview, session, and user information to predict purchasing intention. Other studies also emphasize that purchase prediction in e-commerce is useful for personalization, recommendation systems, and marketing decision-making.

Therefore, this project applies machine learning classification models to predict whether an online shopping session will generate revenue. The main goal is not only to build an accurate predictive model, but also to identify which behavioral factors are most associated with purchase intention and translate the findings into practical business insights for e-commerce decision-making.

1.2 Research Question

Which machine learning classification model best predicts whether an online shopping session will result in a purchase?

1.3 Objectives

The main objective of this project is to develop a machine learning classification model that predicts whether an online shopping session will result in revenue.

Specifically, the project aims to:

1. Analyze customer browsing behavior using session-level features such as page visits, duration, bounce rate, exit rate, page value, traffic type, visitor type, and visit timing.
2. Build and compare several classification models, including Logistic Regression, K-Nearest Neighbors, Random Forest, and MLP Neural Network.
3. Evaluate model performance using F1-Macro instead of accuracy, because the target variable is imbalanced and the business needs to identify revenue-generating sessions more effectively.
4. Select the best-performing model and interpret important features that influence purchasing intention.
5. Provide practical business insights that can support targeted marketing, website optimization, customer engagement, and conversion improvement.

2. Data collection: Describe the dataset, its source, and key characteristics.

2.1 Dataset Source and Structure

The dataset used in this project is the Online Shoppers Purchasing Intention Dataset, sourced from the UCI Machine Learning Repository (Sakar & Kastro, 2018). It contains 12,330 rows and 18 columns, where each row represents a unique user session from a real e-commerce website. The dataset spans 10 months (February through December, excluding January and April), meaning findings may not fully represent seasonal patterns across a complete calendar year. The dataset was deliberately structured so that each session belongs to a different user, avoiding bias from repeat visits or specific campaign periods.

Data quality check results:

- No missing values detected across any column
- 125 duplicate rows found

2.2 Feature Description

Table 1: Dataset Feature Description

Feature	Type	Description
Administrative	Numeric	Number of administrative pages visited
Administrative_Duration	Numeric	Time spent on administrative pages (seconds)
Informational	Numeric	Number of informational pages visited
Informational_Duration	Numeric	Time spent on informational pages (seconds)
ProductRelated	Numeric	Number of product-related pages visited
ProductRelated_Duration	Numeric	Time spent on product-related pages (seconds)
BounceRates	Numeric	Percentage of sessions with no interaction
ExitRates	Numeric	Percentage of pageviews that ended the session
PageValues	Numeric	Estimated value of pages visited before purchase
SpecialDay	Numeric	Closeness to a special day (0–1)
Month	Categorical	Month of the session
OperatingSystems	Numeric	Visitor's operating system (encoded)
Browser	Numeric	Visitor's browser (encoded)

Region	Numeric	Geographic region (encoded)
TrafficType	Numeric	Traffic source (encoded)
VisitorType	Categorical	Visitor type (Returning, New, Other)
Weekend	Boolean	Session occurred on a weekend
Revenue	Boolean (Target)	Whether the session resulted in a purchase

3. Exploratory Data Analysis: Present the main findings and insights from data exploration.

All EDA was conducted on the training set only (80% of data) to prevent data leakage from the test set.

3.1 Class Imbalance - The Central Challenge

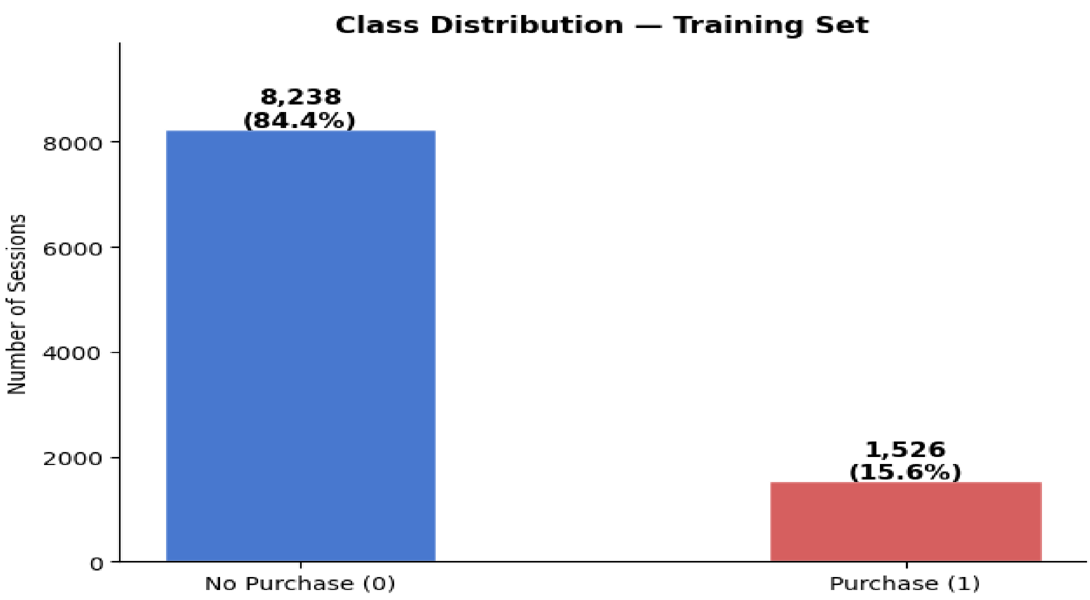

Figure 1: Class Distribution of Purchase and Non-Purchase Sessions

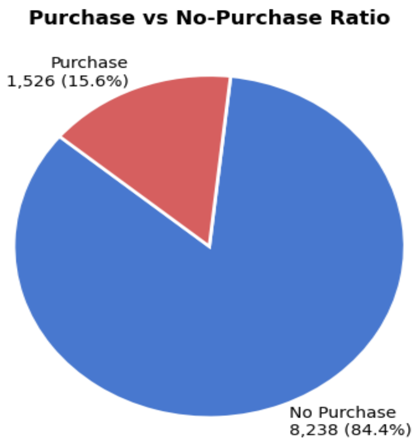

Figure 2: Purchase and Non-Purchase Session Proportion

Only 15.6% of sessions result in a purchase, yielding an approximate 1:5 buyer-to-non-buyer ratio. This is the most consequential finding from EDA. A naive classifier that predicts "no purchase" for every session would already achieve 84.4% accuracy - making accuracy a meaningless metric for this problem. This imbalance directly drove two decisions: the use of *class_weight='balanced'* across applicable models, and the selection of Macro F1 as primary evaluation metric throughout model comparison.

3.2 PageValues Dominates All Other Features

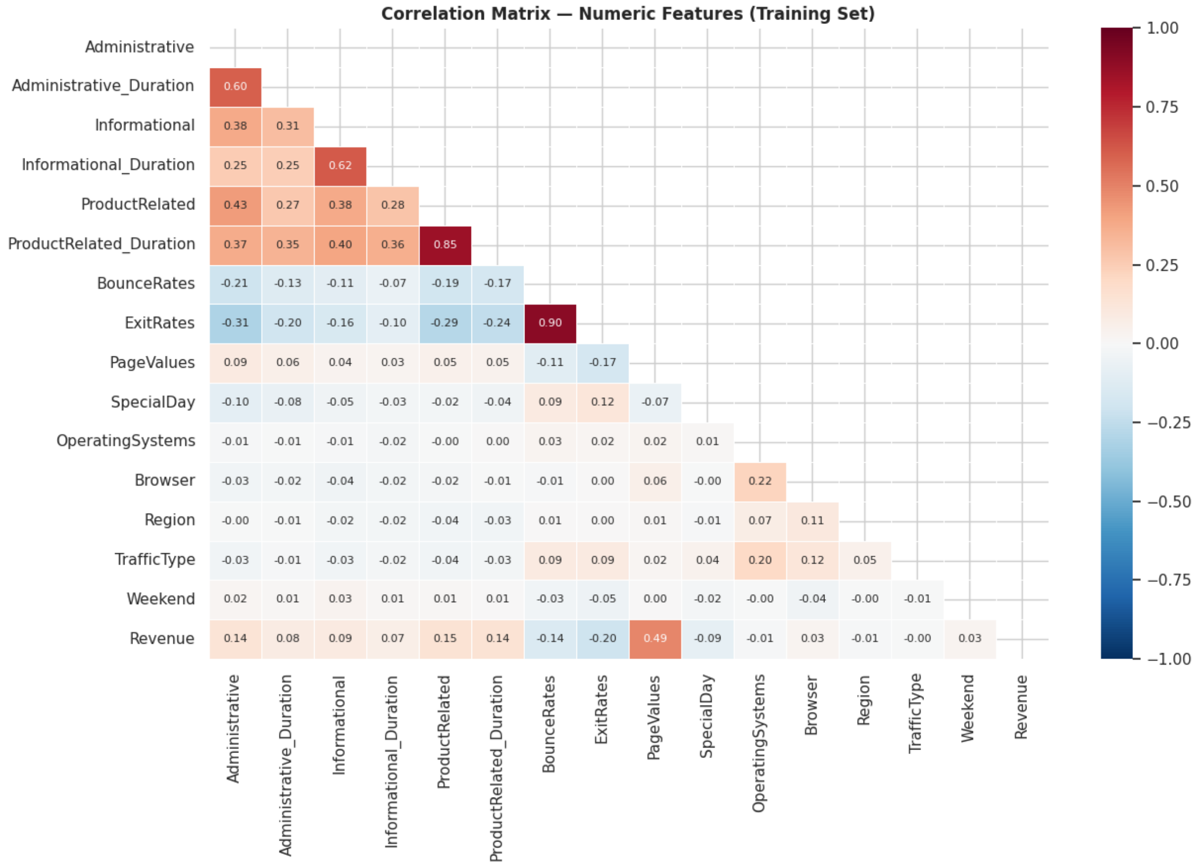

Figure 3: Correlation Matrix of Numeric Features and Revenue

PageValues measures how close a user's current page is to the checkout funnel - cart pages, shipping pages, and payment pages all carry high PageValues. Its correlation with Revenue is **+0.49**, more than double that of any other feature. Critically, this signal is detectable in real time as a session unfolds, making it directly actionable for live intervention systems. The remaining features show weaker individual correlations, suggesting the model must leverage feature combinations rather than any single signal to discriminate effectively.

3.3 Disengagement Behavior and Multicollinearity

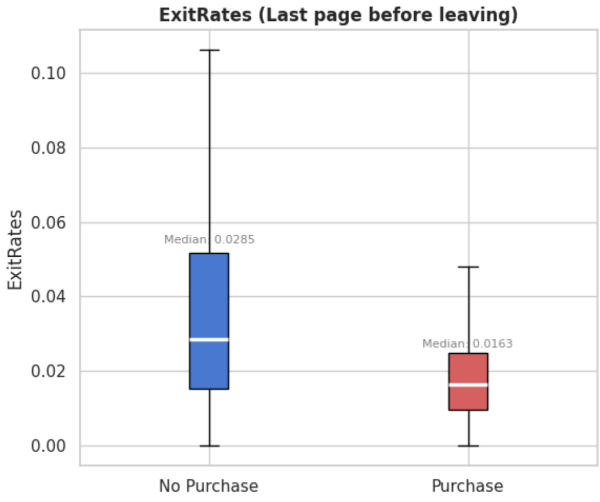

Figure 4: Distribution of Exit Rates by Purchase Outcome

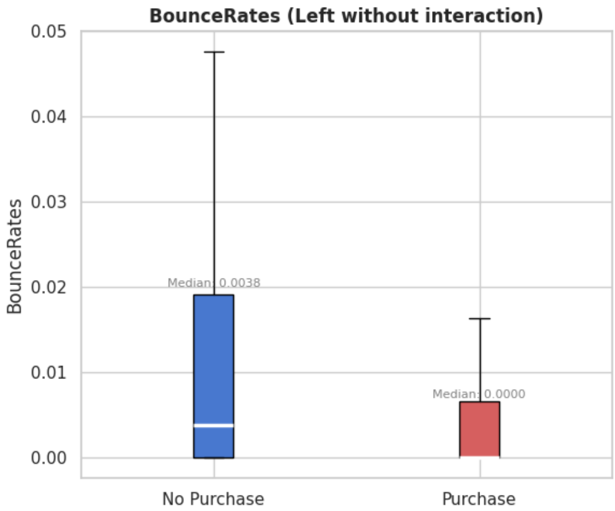

Figure 5: Distribution of Bounce Rates by Purchase Outcome

Non-converting sessions exhibit substantially higher *BounceRates* and *ExitRates*, indicating that users who leave quickly or exit from key pages are much less likely to make a purchase. These two variables are highly correlated (r = 0.90), suggesting they capture similar disengagement behavior. Likewise, ProductRelated and *ProductRelated_Duration* show strong correlation (r = 0.85), as users who view more product pages naturally spend more time on them. Such multicollinearity can reduce coefficient stability in Logistic Regression, whereas Random Forest is largely unaffected due to its tree-based structure, providing an early indication that a tree-based model may be more suitable for this dataset.

3.4 Purchase Behavior Across Visitor Types and Visit Timing

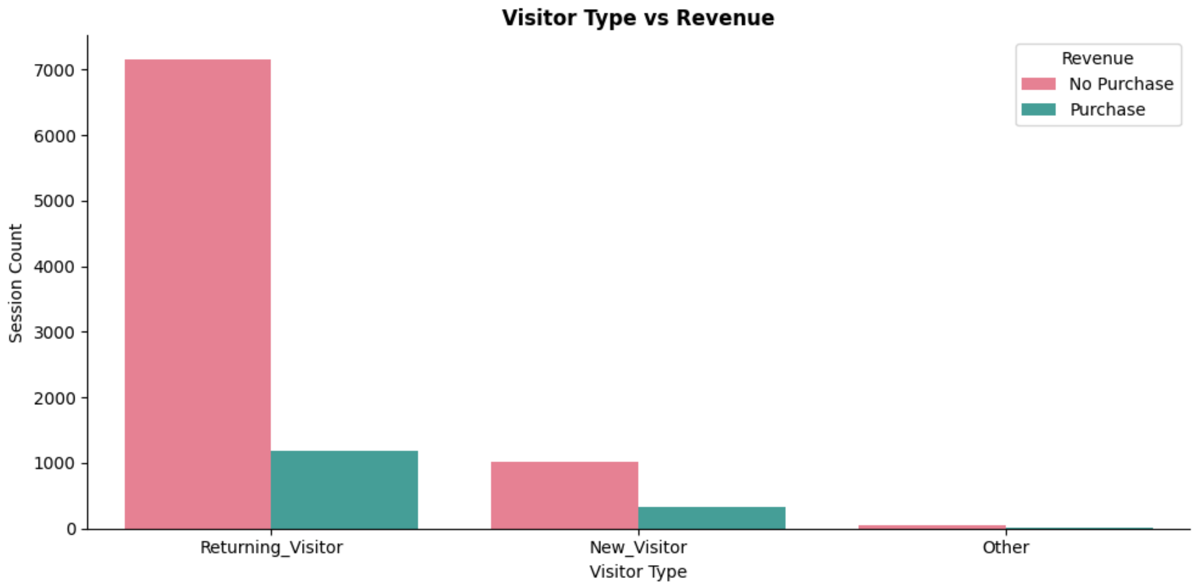

Figure 6: Purchase Behavior Across Visitor Types

Returning visitors account for most purchases, indicating that prior engagement with the website is strongly associated with conversion behavior. This highlights the importance of customer retention as a driver of revenue.

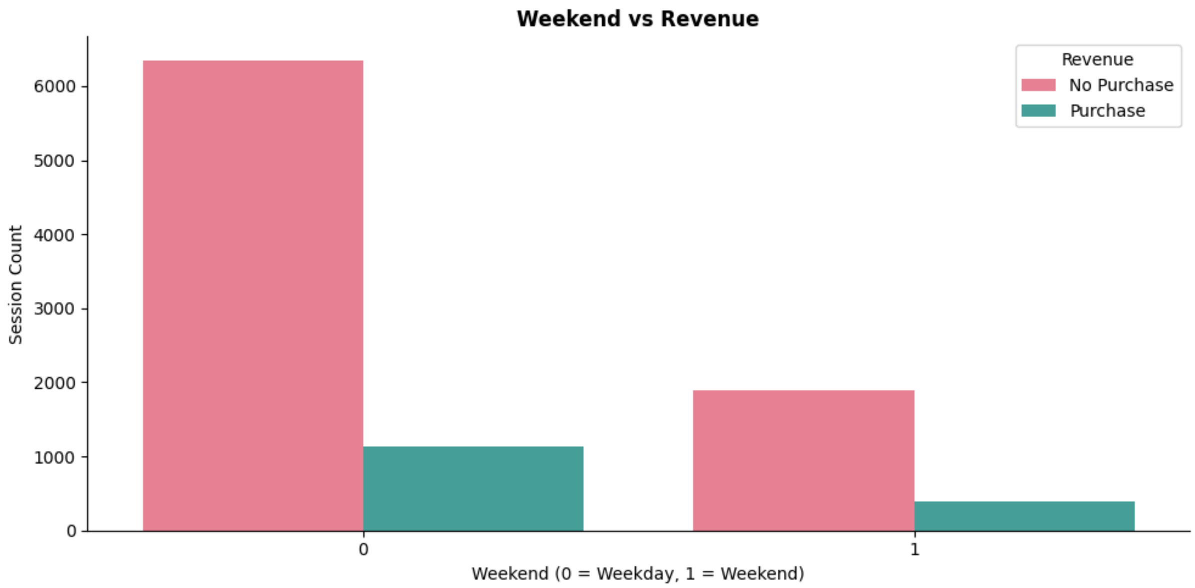

Figure 7: Purchase Behavior Across Weekend and Weekday Sessions

Although weekday sessions account for the majority of website traffic, weekend visitors appear relatively more likely to make a purchase. The proportion of purchase sessions is noticeably higher on weekends compared to weekdays, suggesting that users browsing during weekends may exhibit stronger shopping intent. This indicates that visit timing may provide additional predictive value for distinguishing purchasing and non-purchasing sessions.

3.5 November Leads Both Volume and Conversion Quality

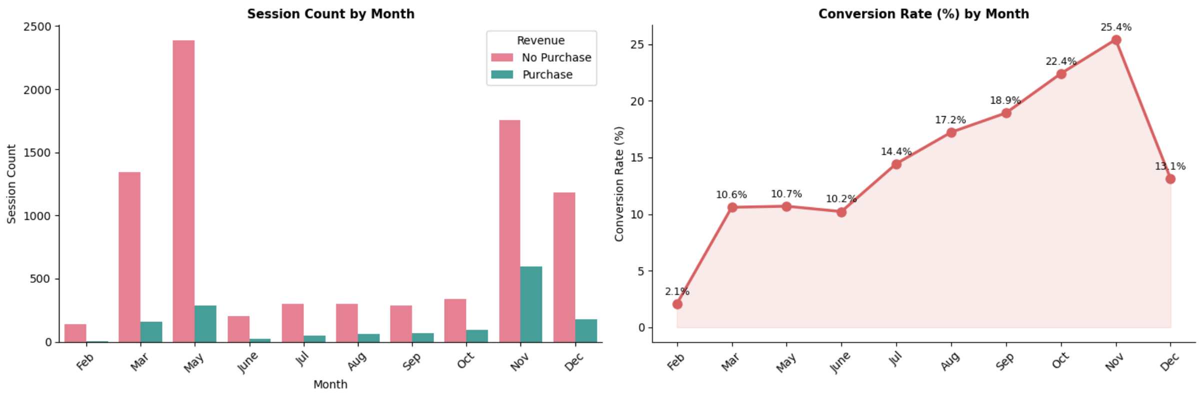

Figure 8: Monthly Session Volume and Conversion Rate

November was the strongest-performing month, achieving both the highest traffic volume and the highest conversion rate (25.4%). Visitors in November were approximately 2.4 times more likely to make a purchase than those in February (2.1%), highlighting a strong seasonal effect in online shopping behavior. Although May attracted the largest number of sessions, its conversion rate remained relatively low (10.7%), suggesting that higher traffic does not necessarily translate into higher sales. Conversion rates generally increased throughout the year, rising steadily from around 10% in the first half of the year to over 25% in November, before dropping sharply to 13.1% in December despite continued high traffic. This pattern suggests that purchase intent was heavily concentrated during major promotional periods, such as Black Friday and Cyber Monday, making the month of visit a potentially valuable predictor of purchase behavior. However, because the dataset covers only ten months and excludes January and April, seasonal patterns should be interpreted with caution.

3.6 Most Visitors Browse Casually - A Few Engage Deeply

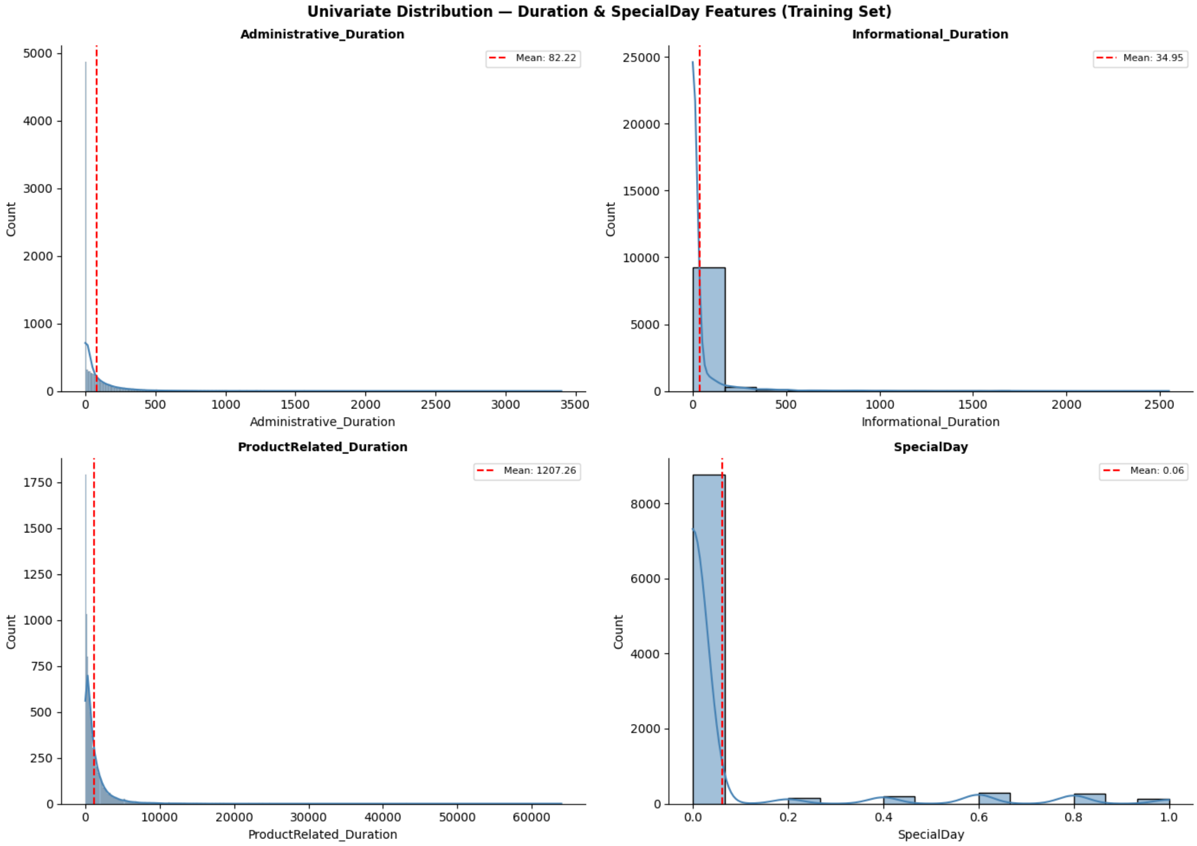

Figure 9: Univariate Distribution of Duration and Special Day Features

All numeric features exhibit heavy right skew that most sessions cluster at low page counts and short durations, while a small number of highly engaged users generate extreme values at the tail. These outliers are not data errors; they represent genuine high-engagement sessions that carry strong predictive signals for purchase behavior. This distribution directly informed two preprocessing decisions: retaining all outliers and applying StandardScaler for distance-sensitive models.

4. Data Cleaning and Transformation: Explain the preprocessing, feature engineering, and rationale behind each step.

4.1 Duplicate Removal

A duplicate check identified 125 duplicated records in the dataset. Since duplicate observations do not provide additional information and may bias the learning process by overrepresenting certain browsing behaviors, all duplicate rows were removed. Retaining duplicates could lead to inflated model performance and reduce the reliability of statistical analyses.

The duplicates were removed before performing the train-test split to prevent identical observations from appearing in both training and testing sets, which could result in data leakage and overly optimistic evaluation metrics.

After removing the 125 duplicate records, the dataset size decreased from (12,330, 18) to (12,205, 18).

4.2 Train/Test Split

The dataset was split before any analysis into 80% training (9,764 sessions) and 20% test (2,441 sessions) using stratified sampling (*stratify=y*, *random_state=42*). Stratification preserves the 15.6% purchase rate in both sets. The test set was locked and not examined until final model evaluation in Section 6.

4.3 Outlier Analysis

Table 2: IQR-Based Outlier Detection Summary

=== OUTLIER ANALYSIS (IQR rule, threshold = 1.5) ===

Feature	Outlier count	% of train
Administrative	325	3.3%
Administrative_Duration	937	9.6%
Informational	2,107	21.6%
Informational_Duration	1,917	19.6%
ProductRelated	767	7.9%
ProductRelated_Duration	751	7.7%
BounceRates	1,127	11.5%
ExitRates	780	8.0%
PageValues	2,187	22.4%
Unique rows flagged (any column)	5,041	51.6%

Using the IQR rule (*threshold = 1.5*), outliers were detected across all numeric features. Removing all flagged rows would eliminate approximately 52% of the training data - an unacceptable reduction. Critically, many extreme values (e.g., very high *PageValues*, long *ProductRelated_Duration*) represent genuine high-engagement sessions that carry strong predictive signals for purchase behavior. Removing them would bias the model toward low-engagement sessions.

Decision: outliers are retained. Sensitivity is instead managed through:

- StandardScaler - normalizes feature ranges for Logistic Regression, KNN, and MLP

- Random Forest - inherently robust to outliers, as tree splits use rank-based thresholds rather than absolute distances

4.4 Categorical Encoding

Two categorical features required encoding:

- **VisitorType:** One-Hot Encoding with drop='first' to avoid the dummy variable trap, producing two binary columns: *VisitorType_Other* and *VisitorType_Returning_Visitor*. The encoder was fitted on the training set only and applied to the test set via *transform()* to prevent data leakage.
- **Month:** Ordinal Encoding using a manually defined calendar mapping (Feb=2 through Dec=12), preserving the natural temporal order and allowing models to detect seasonal trends without inflating the feature space. Since the mapping was manually specified rather than learned from the data, it was applied consistently to both the training and test sets without requiring a fitting procedure or introducing data leakage.

4.5. Feature Scaling

Following encoding, features were separated into predictor matrices (X) and target vectors (y) for both sets. *StandardScaler* was fitted exclusively on the training X matrix and applied to both train and test sets. Logistic Regression, KNN, and MLP received the scaled data; Random Forest received the unscaled training data directly, as tree-based models are invariant to feature magnitude.

5. Model Development

Four classification models were built and compared using 10-fold cross-validation on the training set: Logistic Regression, K-Nearest Neighbors, Random Forest, and MLP Neural Network. The test set remained locked during this phase and was not used for model training or model selection.

The target variable is imbalanced, with around 84.37% non-purchase sessions and 15.63% purchase sessions in the training set. Therefore, F1-Macro was used as the main evaluation metric because it gives equal importance to both classes. Class weighting was applied where the algorithm supported it.

5.1 Logistic Regression

Logistic Regression was used as the baseline model. It estimates the probability of a purchase using a linear combination of input features and the sigmoid function. The model was configured with max_iter=1000 to support convergence and class_weight='balanced' to reduce bias toward the majority non-purchase class.

Scaled features were used as input because Logistic Regression is sensitive to feature scale. This model is useful for comparison because it is simple and interpretable. However, its main limitation is that it can only model linear relationships, while online shopping behavior may involve more complex patterns.

5.2 K-Nearest Neighbors (KNN)

KNN classifies each session based on the majority class among its nearest neighbors. The number of neighbors was selected using the rule of thumb $k = \sqrt{n\_train}$, rounded to the nearest odd number to avoid ties in binary classification. In this project, k was set to 99.

Scaled features were used because KNN depends on distance calculation. Without scaling, features with larger numeric ranges would dominate the distance measure. A key limitation of KNN is that it does not have a built-in class_weight parameter. Since non-purchase sessions are the majority class, buyer sessions may be outvoted by non-buyers during prediction.

5.3 Random Forest

Random Forest was built as an ensemble model using 200 decision trees. Each tree was trained on a bootstrap sample of the training data, and final predictions were made by majority vote. The model was configured with n_estimators=200, max_depth=15, class_weight='balanced', random_state=42, and n_jobs=-1.

Raw unscaled features were used for Random Forest because tree-based models do not require feature scaling. Random Forest is suitable for this dataset because it can capture non-linear relationships and feature interactions. It is also less sensitive to outliers than distance-based models. In addition, Random Forest provides feature importance scores, which can support business interpretation.

5.4 MLP Neural Network for Classification

MLP Neural Network was included as the deep learning model for classification. Since the target variable Revenue is binary, MLPClassifier was used to predict whether each online shopping session would result in a purchase or not. The model can learn non-linear relationships through hidden layers, making it suitable for comparison with traditional machine learning models.

In this project, the MLP model used two hidden layers with 32 and 16 neurons, ReLU activation, Adam optimizer, alpha=0.001, learning_rate_init=0.001, max_iter=500, early_stopping=True, validation_fraction=0.1, and n_iter_no_change=20. Scaled features were used because MLPClassifier relies on gradient-based optimization and is sensitive to feature scale.

Unlike Logistic Regression and Random Forest, sklearn's MLPClassifier does not directly support class_weight='balanced'. Therefore, no class weighting was applied to this model. F1-Macro was used as the evaluation metric to assess performance across both purchase and non-purchase classes. In this project, MLP was treated as a deep learning benchmark model rather than assumed to be the best approach, because the dataset is structured tabular data.

6. Model Evaluation and Selection

6.1 Evaluation Metric

The models were evaluated using F1-Macro as the main evaluation metric. This metric was selected because the target variable is imbalanced, with most sessions belonging to the non-purchase class. If accuracy is used alone, the result may be misleading because a model can achieve high accuracy by predicting the majority class well while still performing poorly in identifying purchase sessions.

F1-Macro is more suitable for this project because it gives equal importance to both classes by averaging the F1-score of the purchase and non-purchase classes. This fits the business objective of the project, which is not only to classify non-buyers correctly, but also to identify sessions that are likely to generate revenue.

6.2 Cross-Validation Results and Model Selection

All four models were evaluated using 10-fold cross-validation on the training set before the test set was used. F1-Macro was selected as the main metric because the target variable is imbalanced, with 84.37% non-purchase sessions and only 15.63% purchase sessions in the training set.

Logistic Regression achieved a mean F1-Macro of 0.7765 with a standard deviation of 0.0192. It served as a useful baseline model and benefited from feature scaling and balanced class weighting. However, because Logistic Regression is a linear model, it may not fully capture complex relationships between browsing behavior and purchase intention.

K-Nearest Neighbors achieved the lowest result, with a mean F1-Macro of 0.6045 and a standard deviation of 0.0205. This weaker performance is reasonable because KNN depends on distance-based voting. Since non-purchase sessions are the majority class, purchase sessions can easily be outvoted by the larger non-purchase group.

MLP Neural Network was included as the deep learning benchmark and achieved a mean F1-Macro of 0.7669 with a standard deviation of 0.0178. The model used the scaled dataset because neural networks are sensitive to feature scale. Although MLP was able to learn some non-linear patterns, it did not outperform Logistic Regression or Random Forest. This suggests that a neural network was not the strongest approach for this structured tabular dataset.

Random Forest achieved the best performance, with the highest mean F1-Macro of 0.8048 and the lowest standard deviation of 0.0121. This indicates that Random Forest not only performed better than the other models but also produced the most stable results across validation folds. Therefore, Random Forest was selected as the final model. It is suitable for this project because it can capture non-linear feature interactions, handle outliers better than distance-based models, address class imbalance through balanced class weighting, and provide feature importance scores for business interpretation.

6.3 Final Evaluation on Test Set

After Random Forest was selected based on cross-validation, the final model was retrained on the full training set of 9,764 sessions and evaluated once on the locked test set of 2,441 sessions. The test set was not used during preprocessing decisions, cross-validation, or model selection. Therefore, the test result reflects the model’s ability to generalize to unseen data.

On the test set, the final Random Forest model achieved an accuracy of 0.9000. This means that the model correctly classified 90.00% of all sessions. However, because the dataset is imbalanced, accuracy is reported only as a general indicator. The purchase class, represented by Revenue = 1, is more important for business interpretation.

For the purchase class, the model achieved a precision of 0.6825, recall of 0.6754, and F1-score of 0.6789. This means that around 68.25% of the sessions predicted as purchases were actual purchases. It also means that the model correctly identified around 67.54% of all actual buyers in the test set.

The confusion matrix shows that the model correctly identified 258 actual buyers and 1,939 actual non-buyers. It also produced 120 false positives and missed 124 actual buyers. From a business perspective, the false negatives are the most important issue because they represent potential buyers who were not identified by the model. These missed buyers may lead to lost opportunities for targeted promotion, product recommendation, checkout reminders, or customer support.

7. Interpretation and Insights

The final Random Forest model provides useful business insights because it can help the company identify website sessions that are more likely to generate revenue. Instead of treating all visitors the same, the business can use the model as a purchase-intention scoring tool to prioritize customers with stronger buying signals.

The results show that customer browsing behavior is strongly related to purchase intention. Variables such as PageValues, ProductRelated, ProductRelated_Duration, BounceRates, and ExitRates are important indicators of whether a session may lead to a purchase. In practical terms, visitors who reach valuable pages, view more product-related content, and spend more time browsing products are more likely to buy. On the other hand, high bounce rates and exit rates suggest weaker engagement and lower purchase intention.

From a business perspective, the model can support more targeted marketing decisions. Sessions predicted as likely purchases can receive timely actions such as product recommendations, checkout reminders, limited-time offers, free-shipping messages, or live chat support. These actions should be focused on visitors with higher purchase probability rather than applied equally to all users, which can help reduce unnecessary marketing costs.

The model also highlights the importance of reducing missed buyers. False negatives represent actual buyers that the model failed to identify. In business practice, these missed buyers may become lost opportunities because the company does not provide timely intervention. Therefore, if the company's priority is to capture more potential buyers, the prediction threshold can be adjusted to increase recall, although this may also increase false positives and marketing costs.

Based on the findings, the business should focus on three practical actions. First, prioritize high-intent visitors based on model prediction scores and PageValues. Second, improve product pages by strengthening product descriptions, images, reviews, comparison tools, and related-product recommendations. Third, reduce bounce and exit rates by improving landing pages, website speed, navigation flow, and call-to-action placement.

8. Team Member Roles: Provide a table assigning specific tasks to each team member.

The table below outlines the specific tasks assigned to each team member for this project. All members contributed to the final report and presentation.

Task	Description	Assigned To
Business Context	Problem definition, research question, objectives, feature descriptions	Nguyen Ngoc Minh Thu, Nguyen Thi Hoai Thuong
Data Loading & Quality	Load dataset, inspect structure, check for missing values and duplicates, boolean conversion	Tran Ngoc Trung Hieu
EDA - Univariate	Class distribution, distribution plots for categorical and numeric features, box plots	Tran Ngoc Trung Hieu
EDA - Bivariate	Feature vs Revenue analysis, engagement metrics comparison, correlation heatmap	Nguyen Ngoc Minh Thu
Outlier Analysis	IQR-based outlier quantification, skewness analysis, rationale for retaining outliers	Nguyen Thi Hoai Thuong

Data Transformation	Categorical encoding (One-Hot, Ordinal), train/test split, StandardScaler, X/y separation	Tran Ngoc Trung Hieu
Model Development	Build LR, KNN, and Random Forest; 10-fold CV comparison; model selection rationale	Nguyen Ngoc Minh Thu
Final Model & Evaluation	Fit final RF on full training set; confusion matrix; ROC curve; feature importance; performance summary	Nguyen Thi Hoai Thuong
Business Insights & Deployment	Feature interpretation, five actionable recommendations, real-time deployment strategy, KPI monitoring plan	All members
Report Write-Up	Full written report (PDF), proofreading, formatting, references	All Members
Presentation	Prepare presentation slides, oral presentation covering methodology, results, and business recommendations.	All Members

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